

Optimizing Dynamic Pricing Strategies with Metaheuristics: A Simulation-Based Optimization Framework

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Abstract

Revenue management has traditionally focused on determining the optimal allocation of limited capacity through booking limits, bid prices, and dynamic pricing strategies. While classical approaches such as Expected Marginal Seat Revenue (EMSR), network bid-price control, and dynamic programming have achieved considerable success in airline revenue management, they typically assume simplified customer behavior and often neglect strategic waiting decisions made by rational customers.

This whitepaper introduces a simulation-based framework for evaluating competitive pricing strategies in a shared airline market where two airlines operate simultaneously under identical seat capacities but employ different pricing mechanisms. One airline adopts an adaptive dynamic pricing strategy while its competitor maintains a fixed static fare throughout the booking horizon. Passengers arrive according to a Poisson process, belong to heterogeneous market segments, evaluate utilities across multiple alternatives, and may strategically postpone purchasing decisions by entering a spillover queue for future re-evaluation. This rational waiting mechanism represents a key innovation because it removes the structural dominance of immediate purchasing and produces more realistic market dynamics.

Furthermore, the paper proposes a metaheuristic optimization framework, using other heuristics as the primary example, to optimize pricing policy parameters through repeated simulation episodes. The resulting methodology provides a flexible decision-support framework that extends naturally to hospitality, retail, ride-sharing, cloud computing, energy markets, and other industries requiring dynamic pricing under uncertain demand.

1 Introduction

1.1 Background

Revenue Management (RM) is concerned with selling the right product to the right customer at the right time for the right price. Since its adoption by the airline industry in the late 1970s following deregulation, RM has become one of the most influential applications of operations research.

Modern RM systems combine

- demand forecasting,
- inventory control,
- pricing,
- customer segmentation,
- and optimization.

The airline industry remains the canonical example because each flight possesses fixed capacity, highly perishable inventory, stochastic demand, and multiple customer segments exhibiting heterogeneous willingness-to-pay.

Traditional airline RM methods include

- Nested booking limits,
- Expected Marginal Seat Revenue (EMSR),
- Bid-price control,
- Dynamic Programming,
- Approximate Dynamic Programming,
- Choice-based Revenue Management,
- and increasingly, Reinforcement Learning.

Although these approaches have substantially improved airline profitability, many rely upon simplified assumptions regarding customer behavior. Passengers are frequently assumed to accept or reject offers immediately after arrival, while strategic waiting behavior is either ignored or represented only implicitly.

1.2 Motivation

The rapid digitalization of travel markets has fundamentally altered customer decision-making.

Consumers can now

- monitor prices continuously,
- compare competing airlines,
- receive automated price alerts,
- postpone purchases,
- and dynamically revise their booking decisions.

Consequently, purchasing decisions are no longer static. Instead, passengers actively reason about the trade-off between

1. purchasing immediately,
2. waiting for a potentially lower future price,
3. switching to a competitor,
4. or abandoning the purchase altogether.

Most educational pricing simulations simplify this process by assuming that "waiting" is merely another action with an arbitrarily assigned utility. In many implementations, however, waiting becomes structurally dominated because there exists no realistic future mechanism that can compensate for the opportunity cost of postponement.

This work addresses this limitation by introducing a **rational waiting mechanism**. Passengers explicitly evaluate expected future price reductions, remaining seat availability, booking urgency, and opportunity costs before choosing whether to wait. Passengers who postpone purchasing enter a spillover queue and periodically re-evaluate their decisions as market conditions evolve.

This mechanism creates realistic competitive dynamics that emerge naturally rather than being imposed by predefined behavioral rules.

1.3 Simulation Overview

The proposed simulation models a competitive market consisting of two airlines offering identical physical capacity.

- Airline A employs a dynamic pricing strategy.
- Airline B maintains a fixed static fare.

- Both airlines possess identical seat capacities.
- Customers arrive randomly according to a Poisson process.
- Two heterogeneous passenger segments are considered: leisure and business travelers.
- Each arriving passenger observes current market conditions and computes the utility associated with every feasible action.
- Passengers may
 1. purchase from the dynamic airline,
 2. purchase from the static airline,
 3. postpone purchasing,
 4. or leave the market.
- Waiting passengers enter a spillover queue and periodically re-evaluate their decisions.
- Dynamic prices continuously respond to booking pace while static prices remain unchanged.
- Both airlines compete simultaneously for the same customer population.

Unlike many revenue-management demonstrations that optimize a single firm’s pricing policy in isolation, this framework explicitly models competitive interaction between pricing strategies within a shared demand environment.

1.4 Research Contributions

This whitepaper makes several contributions to the literature on pricing simulation and educational serious games.

1. A realistic simulation framework integrating stochastic passenger arrivals, utility-based customer decisions, inventory constraints, dynamic pricing, static pricing, and competitive market interaction.
2. A rational waiting mechanism in which postponement becomes an economically meaningful decision instead of an artificially assigned action.
3. A spillover queue allowing customers to repeatedly revise purchasing decisions as prices and seat availability evolve over time.
4. An extensible architecture suitable for educational serious games, algorithm benchmarking, and optimization research.

5. A metaheuristic optimization framework capable of tuning pricing parameters without requiring analytical derivatives or explicit mathematical models.
6. A demonstration that simulation-based optimization can bridge the gap between classical revenue management methods and modern data-driven pricing systems.

1.5 Positioning Relative to Classical Revenue Management

Classical revenue-management methods generally determine either

- booking limits,
- bid prices,
- or optimal pricing policies.

The proposed simulator is not intended to replace these methods. Instead, it provides a flexible experimental environment in which such algorithms may be evaluated under richer behavioral assumptions.

For example,

- EMSR may determine booking protection levels,
- Bid-price control may determine acceptance thresholds,
- Dynamic Programming may compute optimal inventory values,
- while the simulation evaluates customer responses under bounded rationality and strategic waiting.

Consequently, the framework functions as an experimental platform for comparing classical optimization techniques against adaptive pricing strategies under realistic customer behavior.

1.6 Paper Organization

The remainder of this whitepaper is organized as follows.

Section 2 formally describes the simulation model, including passenger behavior, pricing policies, utility functions, spillover dynamics, and performance metrics.

Section 3 proposes a new Metaheuristic Algorithm for optimizing the dynamic pricing policy and explains how simulation episodes are embedded within the optimization loop.

Section 4 presents the recommended experimental design, benchmark scenarios, and evaluation metrics.

Section 5 discusses expected experimental outcomes and sensitivity analyses.

Section 6 demonstrates how the framework generalizes beyond airlines to hospitality, retail, ride-sharing, cloud computing, event ticketing, and energy markets.

Finally, Section 7 summarizes the principal findings and outlines promising directions for future research, including multi-agent reinforcement learning, approximate dynamic programming, network revenue management, and adaptive pricing under competitive market conditions.

2 Simulation Model

2.1 Overview

The proposed simulator models a competitive airline market in which two airlines simultaneously compete for a common stream of customers.

Unlike conventional revenue management simulators that consider only one supplier, this framework explicitly models competition, heterogeneous passenger preferences, stochastic arrivals, adaptive pricing, inventory constraints, and strategic customer behavior.

The simulator is event-driven and progresses over a finite booking horizon until either

1. the booking horizon expires, or
2. both airlines exhaust their seat capacities.

Each arriving passenger observes the current market state and independently chooses the action that maximizes expected utility.

2.2 System Architecture

The simulator consists of four interacting components:

1. Passenger Arrival Generator
2. Airline Pricing Engine
3. Customer Decision Model
4. Performance Evaluation Module

Figure 1 illustrates the interaction among these components.

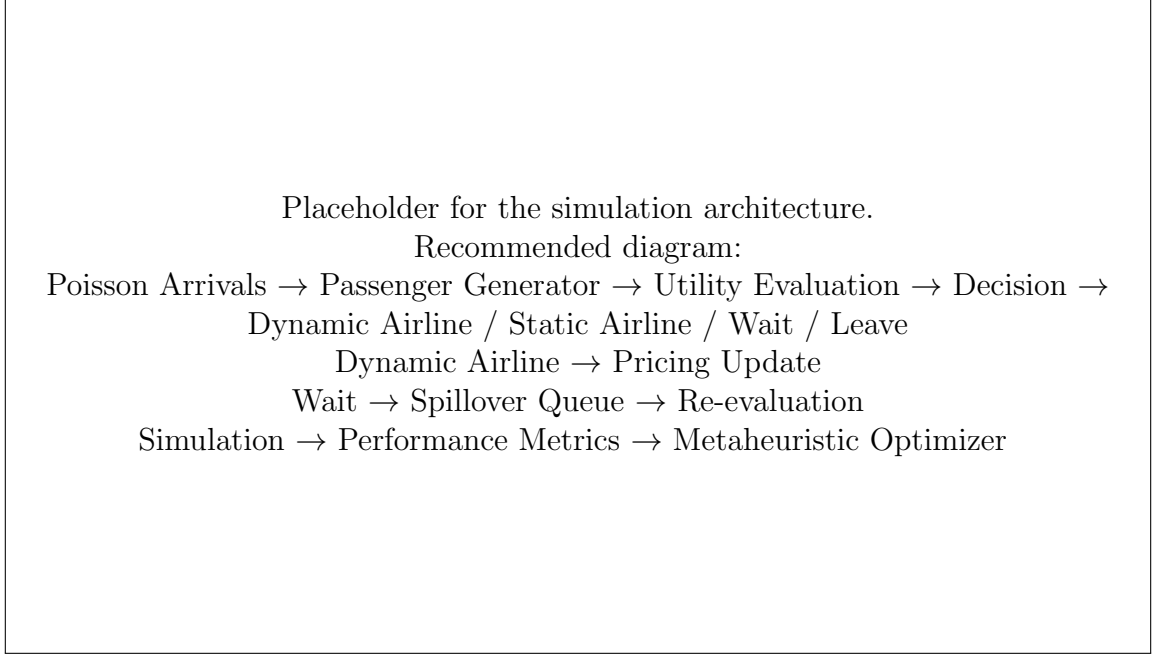


Figure 1: Architecture of the simulation framework and optimization loop.

2.3 Simulation Parameters

Table 1 summarizes the principal parameters used in the baseline implementation.

Table 1: Baseline simulation parameters.

Parameter	Symbol	Baseline
Booking horizon	T	40 time units
Airline capacity	C	20 seats
Total market capacity	$2C$	40 seats
Static ticket price	P_s	\$273
Seat production cost	c	\$200
Arrival rate	λ	0.5–2.5
Pricing coefficient	α	0.618–2.718
Passenger types	–	Leisure / Business
Arrival process	–	Poisson
Decision rule	–	Utility maximization

2.4 Passenger Arrival Process

Customer arrivals follow a homogeneous Poisson process.

Let

$$N(t)$$

denote the cumulative number of arrivals before time t .

The arrival process satisfies

$$N(t) \sim \text{Poisson}(\lambda t), \quad (1)$$

where

$$\lambda > 0$$

represents the expected arrival rate.

Inter-arrival times therefore follow the exponential distribution

$$\Delta t \sim \text{Exp}(\lambda), \quad (2)$$

with probability density

$$f(\Delta t) = \lambda e^{-\lambda \Delta t}. \quad (3)$$

This stochastic arrival process captures the uncertainty observed in real-world booking environments while remaining computationally efficient.

2.5 Passenger Segmentation

Passengers are divided into two behavioral classes.

1. Leisure travelers
2. Business travelers

Each passenger possesses an individual willingness-to-pay

$$v_i.$$

Business travelers exhibit

- higher urgency,
- higher willingness-to-pay,
- lower sensitivity to waiting,
- and greater tolerance for expensive tickets.

Leisure travelers generally display the opposite characteristics.

This segmentation creates heterogeneous purchasing behavior without requiring high-dimensional customer models.

2.6 Dynamic Pricing Model

Only Airline A updates ticket prices dynamically.

The dynamic fare is determined from the booking pace relative to the expected booking trajectory.

Let

$$B(t)$$

represent cumulative bookings of the dynamic airline.

Expected bookings are

$$E(t) = C \cdot \frac{t}{T}. \quad (4)$$

The booking pace ratio becomes

$$R(t) = \frac{B(t)}{\max\{1, E(t)\}}. \quad (5)$$

The dynamic price is computed as

$$P_d(t) = P_s R(t) \alpha, \quad (6)$$

where

- P_s is the static fare,
- α is the pricing coefficient.

Price limits prevent unrealistic behavior,

$$P_{\min} \leq P_d(t) \leq P_{\max}. \quad (7)$$

Consequently,

- bookings above expectation increase prices,
- weak demand lowers prices,
- inventory scarcity naturally increases ticket values.

2.7 Static Pricing Model

Airline B follows a conventional fixed-price policy.

Its fare remains

$$P_s = \text{constant} \quad (8)$$

throughout the booking horizon.

No inventory-based adjustments are performed.

Although simplistic, this airline provides an interpretable benchmark against which adaptive pricing strategies can be evaluated.

2.8 Utility-Based Customer Decisions

Each passenger evaluates four possible actions:

1. Buy from Dynamic Airline
2. Buy from Static Airline
3. Wait
4. Leave

Passengers choose the action with maximum expected utility.

2.8.1 Buying from Dynamic Airline

Utility equals

$$U_D = V_i - P_d + Q_D, \tag{9}$$

where

- V_i denotes passenger valuation,
- P_d denotes the current dynamic fare,
- Q_D represents seat-quality utility.

2.8.2 Buying from Static Airline

Similarly,

$$U_S = V_i - P_s + Q_S. \tag{10}$$

2.8.3 Leaving the Market

Leaving generates

$$U_L = 0. \tag{11}$$

This normalization permits utilities to be interpreted relative to the outside option.

2.9 Rational Waiting Model

One of the primary innovations of this simulation is the rational waiting mechanism.

Rather than assigning an arbitrary waiting utility, passengers estimate the potential value of postponing their purchase.

Waiting utility is computed as

$$U_W = B - C_u - C_t, \quad (12)$$

where

- B is the expected future benefit,
- C_u is urgency cost,
- C_t represents elapsed-time penalty.

Expected benefit depends upon

- remaining seat inventory,
- remaining booking time,
- current price premium,
- expected future price reductions.

An example formulation is

$$B = \gamma S_r T_r \Delta P V_i, \quad (13)$$

where

- S_r is remaining seat ratio,
- T_r is remaining time ratio,
- ΔP is current price premium,
- γ is a calibration coefficient.

Unlike many educational pricing games, waiting is therefore not artificially penalized. Instead, passengers wait only when expected future gains exceed the costs of delay.

2.10 Spillover Queue

Passengers selecting the waiting action enter a spillover queue.

At regular intervals they

1. observe updated prices,
2. inspect remaining seat inventory,
3. recompute utilities,
4. either purchase,
5. continue waiting,
6. or leave permanently.

This repeated decision process transforms the booking market into a dynamic system with endogenous demand feedback.

Figure 2 illustrates the conceptual process.

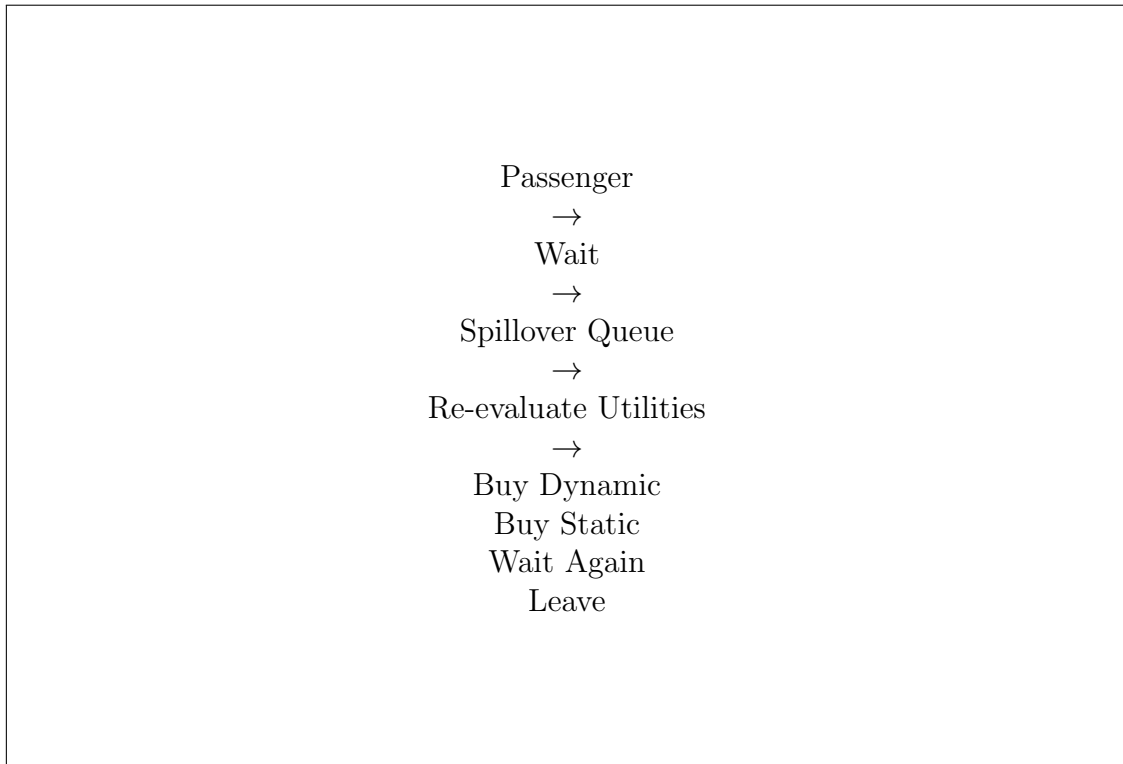


Figure 2: Repeated spillover evaluation process.

2.11 Seat Inventory

Each airline owns

$$C = 20$$

available seats.

Once sold,

inventory cannot be replenished.

Seat allocation follows

$$B(t) \leq C. \quad (14)$$

When capacity reaches zero,

the airline exits the booking market while competitors continue serving remaining demand.

2.12 Performance Metrics

Several key performance indicators are recorded throughout every simulation episode.

Table 2: Performance metrics collected during simulation.

Metric	Description
Revenue	Total ticket income
Profit	Revenue minus production cost
PnL%	Profit margin percentage
Market share	Fraction of passengers served
Load factor	Seat utilization
Average ticket price	Mean selling price
Customer satisfaction	Utility-based score
Average waiting time	Spillover delay
Spillover rate	Waiting passengers / arrivals
Lost demand	Customers leaving market

Profit margin is computed as

$$PnL(\%) = 100 \times \frac{\text{Revenue} - \text{Cost}}{\text{Cost}}. \quad (15)$$

2.13 Simulation Flow

The complete simulation proceeds according to the following sequence:

1. Initialize airline inventories.
2. Generate passenger arrivals.
3. Update the dynamic ticket price.
4. Process newly arriving passengers.
5. Evaluate utilities.

6. Execute purchase, waiting, or exit decisions.
7. Re-evaluate spillover passengers.
8. Record performance metrics.
9. Continue until booking horizon ends or all seats are sold.

The resulting framework provides a realistic yet computationally efficient environment for evaluating pricing policies under stochastic demand, competition, bounded rationality, and adaptive customer behavior.